2A H&E → DM1 image projection — negative-feasibility finding

Tile-level H&E prediction of depth-residualized 8-gene DM1 score is bounded below random – a pre-registered failed hypothesis as methods supplement. 다음 스튜디오에 "이 *dataset* 으로 같은 시도 하지 말라"는 *actionable upper bound* 를 제공한다.

NO-GO 확정 2026-05-04

~85 min CPU + RunPod ~$30 sunk

Re-entry 5/5 conditions 必

Author: **Seungho Cook**

Anchor: **Paper 2 H&E projection · methods supplement**

Date: 2026-05-04

I. 한 줄 결론

핵심: ResNet50 224 px LOSO 의 DM1_resid Spearman r = 0.022 — 50 random gene panel 의 max (r = 0.062) 보다 낮음. **Foundation model 업그레이드 projected gain (+0.05 ~ +0.15) 도 GO threshold (+0.28) 미달.** Verdict D (full 폐기).

DM1_RESID LOSO SPEARMAN	RANDOM PANEL MAX (RAW)	FOUNDATION PROJECTED GAIN	RE-ENTRY CONDITIONS
0.022	**0.062**	**+0.15**	**0/5**
ResNet50 224 px · chance (≤ random panel max)	50 random 8-gene panels · real RAI raw 0.067 ≈ random	UNI/ CONCH/ Virchow2 — still below GO +0.28	All five must clear before any retry

PROBLEM

Visium hires (~50 μm) spot-level molecular label vs tile-level H&E content 의 fundamental mismatch.

IDEA

Closure battery (5-failure-mode) 로 mismatch 를 정량적으로 BOUND. Verdict D 도출.

DEFENSE

raw 9-dim RGB r=0.224 >> ResNet50 0.061 → raw 신호의 정체는 **stain↔depth artifact**. residualization 정당.

DECISION

Image-DM1 paper axis dropped. Methods supplement 으로 박제 + Paper 7 Pattern 4 case study.

이 NO-GO 가 알려주는 3 가지 핵심 boundary:

1. **spot-level ST molecular label ↔ tile-level H&E content 의 fundamental mismatch** at hires Visium (~50 μm). 이건 **resolution-specific limitation** — full-res WSI 에서는 다를 수 있음.

2. **raw signal 의 정체는 stain↔depth artifact** (raw 9-dim RGB r=0.224 >> ResNet50 0.061). residualization 정당 — depth confounder 차단.
3. **Foundation model 만으로는 못 풂.** UNI/CONCH/Virchow2 projected gain +0.05~+0.15 인데 GO threshold +0.28 미달. (1)+(2) 가 dominant.

FAILURE → FINDING. Visium ST 의 spot-level molecular label 과 tile-level H&E content 사이에 **fundamental mismatch** 가 존재한다는 것을 정량적으로 BOUND 했다. ResNet50 ImageNet 224 px DM1_resid Spearman r = 0.022 (chance). 50 random gene panels max = 0.062 — real RAI_8 raw = 0.067 ≈ random max. → Methods supplement / pre-registered failed hypothesis 로 **negative result publication 가치**.

2. 비전공자 배경

Visium spatial transcriptomics (Visium) 는 **한 spot 마다 (한 spot ≈ 50 µm × 50 µm)** 의 mRNA 측정 + 같은 spot 위에 매칭된 **H&E 영상 tile (224 × 224 px)** 을 동시에 보유. 즉 분자 (RNA) 와 형태 (H&E) 가 **1:1 spatial alignment** 로 paired. **가설:** H&E tile 의 morphology 만 보고 → 그 spot 의 8-gene RAI 점수 (depth-residualized DM1_like_score_resid) 를 예측 가능하다. 이게 됐으면 routine H&E (sequencing 없이) 만으로 RAI loss 를 진단 가능 – 임상 application 의 dream scenario.

2.1 용어 정리

TERM	풀어쓴 설명	PAPER 2A 에서의 역할
Visium ST	10x Genomics 의 spatial transcriptomics 플랫폼. ~50 μm spot 단위로 RNA 와 H&E 동시 측정.	입력 데이터 모달리티. tile-image-to-RNA-axis 매핑의 기반.
DM1_like_score_resid	8 개 thyroid-specific gene (TPO/DIO1/TSHR/PAX8/TG/FOXE1/NKX2-1/SLC5A5) 의 depth-residualized score. Driver excluded.	예측 target. Paper 1 의 핵심 axis.
LOSO	Leave-one-slide-out cross-validation. 16 slide 중 1 개를 held-out 으로.	train/test split. 모델 generalization 측정.
Spearman r	rank-correlation. 비선형 관계도 잡음.	주 metric. r ≥ 0.30 = GO, ≥ 0.20 = BORDERLINE, < 0.20 = NO-GO.
Random panel	≥30% detection rate 인 gene 중 random 으로 8 개 뽑은 control panel.	noise floor 측정. real ≤ random max 면 NO-GO.
Stain↔depth artifact	H&E 진하게 염색된 spot 에서 RNA 도 더 많이 잡히는 spurious correlation.	raw 신호의 confounder. residualization 의 정당화 근거.

3. 전체 논리 흐름

이 페이지는 image-to-DM1 가설을 **5 가지 실패 가능 원인** 으로 가정하고 하나씩 disprove 한 closure battery 의 결과 페이지이다. 결론: 이 dataset (Visium hires) + 이 task (depth-residualized DM1 axis prediction) 의 조합은 안 풀린다 — Verdict D (full 폐기).

1. **가설을 고정한다.** Paper 1 의 depth-residualized 8-gene DM1_like_score_resid 를 H&E tile 로 spot-level prediction.

2. **5-failure-mode 를 정의한다.** 모델 약함 / tile size / residualization / spot-level mismatch / foundation model.

3. **Phase A–G 로 disprove.** 각 phase 가 한 가설을 차단 또는 BOUND.

4. **Random panel 과 비교.** real ≤ noise floor 면 NO-GO.

5. **Foundation model gain projection.** +0.05~+0.15 도 GO threshold +0.28 미달.

6. **Verdict D 도출 + methods supplement 로 박제.** Paper 7 Pattern 4 case study 로 영구 보존.

3.1 분석 사고 스토리

Q1 — 모델이 약해서?

ResNet50 ImageNet pretrained 가 그래도 약하지 않은 baseline. 또 raw 9-dim RGB 가 ResNet50 (r=0.061) 보다 더 잘 (r=0.224) → 모델 약함은 아닌 듯.

Q2 — TILE SIZE 가 작아서?

224 → 448 swap 시 r 0.022 → 0.017 (worse). 672 cancelled. Scale 도 핵심 아님.

Q3 — DEPTH-RESIDUALIZATION 이 MORPHOLOGY-VISIBLE SIGNAL 을 제거?

raw 9-dim RGB r=0.224 vs resid r=0.080. raw 신호의 정체는 **stain darkness ↔ sequencing depth artifact**. residualization 정당.

Q4 — SPOT-LEVEL ST LABEL 과 H&E TILE CONTENT 의 FUNDAMENTAL MISMATCH?

Stage ordinal r = −0.05, ATC binary AUROC = 0.689 (BORDERLINE) — gross morphology 만 잡힘. **분자 단위 신호는 H&E tile 에 fundamentally encoded 안 되어 있음** at hires Visium resolution.

Q5 — FOUNDATION MODEL 업그레이드?

UNI/CONCH/Virchow2 projected gain (+0.05~+0.15) 도 GO threshold (+0.28) 미달. (3)+(4) dominant 라서 모델 강화로 못 풂.

FINAL — VERDICT D (FULL 폐기).

이 dataset + 이 task 조합은 안 풀린다. Methods supplement 로 박제, image-axis 는 Paper 1/2 에서 dropped.

4. Hypothesis (rejected)

Paper 1 의 depth-residualized 8-gene `DM1_like_score_resid` can be predicted from spot-aligned H&E tiles in Visium ST data (28 slides, GSE250521 + 4 GSE230424 + 8 GSE248205).

ITEM	VALUE
Status	NO-GO 확정 2026-05-04 (closure battery, ~85 min CPU + RunPod ~$30 sunk cost)
Pivot	Paper 2B (Hashimoto-overlap mechanism, molecular-only) 로 scope 이동 — image-axis dropped from Paper 1/2 entirely
Re-entry	5/5 conditions 모두 충족 시 only (현재 0/5)
Memory	v52_lodo_finding (관련) · "고고"/"faster"/"다 해줘" override NOT 적용

5. Closure battery — 5-failure-mode dissection

PHASE	TEST	OUTCOME	WHAT IT BOUNDS
A	raw DM1/RAI/TDS @ 224 px ResNet50 (6 metrics × 2 models)	r ≤ 0.08	baseline molecular target weak
B	5 residualization variants (log_counts / log_ngenes / no_resid / rank_norm / epi-top50)	all r ≤ 0.06	residualization choice 도 못 살림
C	tile size 448 px sweep	r = 0.017 (worse than 224)	scale 도 핵심 아님
D	stage ordinal / ATC binary / epithelial / proliferation	stage r=−0.05 / ATC AUROC 0.689 BORDERLINE	ATC만 잡힘 (gross anaplastic morphology, prior art 풍부)
E	slide-level aggregation (mean / top25 / highrisk_frac)	top25_mean 0.209	aggregation 으로도 floor 못 넘김
F	simple RGB 9-dim baseline vs ResNet50	raw DM1 r=0.224 >> ResNet50 0.061	**stain↔depth artifact** 가 raw 신호의 정체 – morphology 아님
G	HF foundation model access	blocked (RunPod stuck)	upgrade 가능성도 +0.05–0.15 만 – threshold 미달
neg ctrl	30 random panels (resid) + 50 random (raw) + housekeeping	real ≤ random max 양쪽 mode	real 신호가 random 대비 우위 없음

Closure battery composite

CLOSURE BATTERY COMPOSITE — 모든 molecular target r ≤ 0.08; ATC binary 만 BORDERLINE 0.689 (anaplastic gross morphology 의 self-evident 신호). residualization은 정당 (stain artifact 차단), 그러나 그 결과 **SPOT-LEVEL ST LABEL과 TILE CONTENT는 FUNDAMENTALLY MISMATCHED.**

pred vs obs

PRED VS OBS RESNET50. observation 과 prediction 의 scatter. 실측 분포가 prediction 에 의해 분리되지 않음 (random distribution 과 구별 불가).

pred vs obs per slide

PRED VS OBS PER SLIDE. slide-by-slide breakdown. 어떤 slide 에서도 consistent prediction signal 없음.

6. Dominant failure cause (verdict D = full 폐기)

1. Depth-residualization removes morphology-visible signal → 신호 자체가 stain↔depth artifact 였음 (residualization 정당화)
2. **Spot-level ST molecular label과 tile-level H&E 의 본질적 mismatch** at hires-Visium (224–672 px) resolution. Foundation model upgrade 도 (1)+(2) 못 풂.

7. Methods supplement 의 자산 가치

이 NO-GO 는 다음 스튜디오에게 **"같은 dataset 으로 같은 시도 하지 말라"** 라는 actionable upper bound 를 제공한다.

Paper 1 / Paper 2B Methods Supplement 에 *pre-registered failed hypothesis* 한 페이지로 박제:

> "We pre-tested H&E-based prediction of the depth-residualized DM1 axis at GSE250521 hires Visium resolution and found no learnable signal beyond random panels (closure battery, Methods Supplementary X)."

→ 이건 **negative result publication** 의 가치. Paper 7 (v18 agentic_research framework) Pattern 4 = "Pre-registered failed hypothesis as methods supplement" 의 case study 로 채택.

7.1 왜 이 NO-GO 가 "발견" 인가 (negative result publication)

- "Visium hires (~50 μm spot) 에서 H&E tile 만으로 spot-level transcriptional axis 를 predict 하려는 시도는 sample-level 에서 정량화된 r ≤ 0.08 의 floor 가 있다."
- "Foundation model upgrade 만으로 이 floor 를 못 넘긴다 (projected +0.05 ~ +0.15 < threshold +0.28)."
- "Stain↔depth artifact 가 raw signal 의 dominant 원인 — residualization 정당."
- "ATC vs non-ATC binary 만 H&E tile 로 잡힘 — anaplastic morphology 의 self-evident 신호이지 새 발견 아님."

8. Re-entry conditions (5/5 모두 필요)

1. Full-resolution scanner WSI for ≥1 cohort + paired bulk RNA-seq
2. External molecular labels (not ST surrogate)
3. Post-marathon explicit decision (after Paper 1 bioRxiv 6/13)
4. Foundation model access (UNI / CONCH / Virchow2) acquired + validated on non-thyroid benchmark
5. Pre-registered hypothesis for re-test

> 현재 0/5. Until all five hold: **no retry.** "TCGA WSI download" + "RunPod G1/G2/G3" 모두 차단.

8.1 Re-entry 5/5 conditions — 현재 상태

#	CONDITION	CURRENT	WHAT'S NEEDED
1	Full-resolution scanner WSI cohort + paired bulk RNA-seq	0/1	40× / 80× WSI cohort + RNA-seq 매칭 (Visium 아닌 진정한 WSI)
2	External molecular labels (not ST surrogate)	0/1	RNA-seq matched bulk label (TCGA WSI 다운로드 금지 상태)
3	Post-marathon explicit decision (after Paper 1 bioRxiv)	pre-marathon	2026-06-13 후
4	Foundation model access (UNI / CONCH / Virchow2) + non-thyroid 벤치 검증	0/1	HF gated access + 다른 cancer type 에서 model performance 검증
5	Pre-registered hypothesis	0/1	OSF / AsPredicted 에 hypothesis + analysis plan 등록 (post-hoc analysis 차단)

9. What this kills

- Image-DM1 Paper 2 active pillar (replaced by Paper 2B)
- TCGA WSI Phase C external validation (~500 GB download blocked)
- Multi-cohort LODO Phase B for image-DM1 purpose
- RunPod / Azure GPU spend toward image-axis
- HF UNI / CONCH / Virchow2 foundation model access for this purpose
- Paper 2 H&E IP claim (color artifact unpatentable; ATC classifier prior art dense)

10. 시각 자료 — closure battery + spatial

10.1 Closure battery 전체 그림

closure battery summary

F1 — CLOSURE BATTERY COMPOSITE. 모든 6 phase + 31 negative-control panel 의 한 줄 비교. real DM1_resid (0.022) ≤ random panel max (0.031); raw RAI (0.067) ≈ random raw max (0.062); 9-dim RGB (0.224) >> ResNet50 (0.061) on raw → stain↔depth artifact 증명.

pred vs obs scatter

F2 — POOLED 224 PX LOSO SCATTER. ResNet50 ImageNet 16-fold leave-one-slide-out, 3,200 tile predictions. Spearman r = 0.022, AUROC = 0.511 (chance).

per-slide LOSO bar

F3 — 16-SLIDE PER-FOLD SPEARMAN. best fold = +0.118 (PTC-1, PTC-4); worst = −0.074 (ATC-3). 16/16 fold 모두 noise level.

10.2 GSE250521 spatial — stage representative slides

N-1 spatial

STAGE 1 / PT (N-1). RAI high · DM1 low · TDS high (well-differentiated). H&E 에서 follicle architecture만 보임.

PTC-1 spatial

STAGE 2 / PTC (PTC-1). Mixed RAI + intermediate DM1. Papillary architecture + nuclear features. Tile-level 에서 DM1_resid mismatch.

LPTC-1 spatial

STAGE 3 / LPTC (LPTC-1). Late-stage. Mixed RAI + elevated DM1. Invasive front + heterogeneous morphology.

ATC-1 spatial

STAGE 4 / ATC (ATC-1). RAI low · DM1 high · anaplastic morphology. 이 stage 만 ATC-vs-non-ATC binary classifier (AUROC 0.689) 가 잡음 – gross histology level.

10.3 GSE250521 sample-mean trends

sample means

F4 — 16-SLIDE SAMPLE-MEAN RAI/DM1/TDS BY STAGE. PT → ATC trajectory 에서 sample-level 신호는 명확하나 spot-level mismatch – closure battery dominant cause (4) 의 시각적 증거.

violin primary

F5 — PER-STAGE SPOT-LEVEL VIOLIN. 각 stage 안의 큰 spread → spot-level heterogeneity 가 H&E feature 와 매핑 안 됨.

violin epi50

F6 — EPITHELIAL-TOP-50% SUBSET. Stromal noise 제거 후에도 ResNet50 가 잡지 못함. Subset filtering 도 image-DM1 rescue 못함.

11. GigaTIME positioning (manuscript-use-only utility)

Recent multimodal AI reference (Valanarasu et al., *Cell* 189[2]:386, 2026, DOI `10.1016/j.cell.2025.11.016`) is positioned as **task-class boundary / future-work comparator only** – NOT competitor, NOT THCA-specific claim, NOT main-text material. THCA inclusion in their 24 cancers is NOT confirmed in public metadata; 21-protein panel composition NOT confirmed; code/weights NOT located as of 2026-05-04 search.

DIMENSION	OUR PAPER 2A TEST	GIGATIME (VALANARASU 2026)
Task class	continuous transcriptional regression	image-to-image protein translation
Output modality	scalar DM1_resid per spot	multi-channel virtual mIF per cell
Training paired data	None for predictor (frozen ImageNet)	~40×10^6 paired cells (21 proteins)
Cohort scale	16 ST slides × 3,200 tiles	14,256 patients × 51 hospitals × 24 cancers
Cancer scope	thyroid only (PT → ATC)	24 types, 306 subtypes (THCA inclusion unknown)
Result	r = 0.022, AUROC = 0.511 (random)	1,234 protein-biomarker-staging-survival associations
Pre-registered gate	NOT met (Spearman < 0.20 BORDERLINE)	not applicable (different task)

→ complementary, not competing. GigaTIME demonstrates H&E carries signal when paired with molecular ground-truth at scale; our test bounds H&E-alone recoverability for transcriptional axes. Bridging requires RNA-paired thyroid training at $\geq 10^2$ patients full-res WSI (re-entry condition #1+#2).

12. Reviewer Q1–Q8 condensed (Q9 voice-protected, EXCLUDED)

Q	LIKELY TOPIC	ANSWER KEY (≤2 SENTENCES)
Q1	Why frozen ResNet50, not UNI/CONCH/ Virchow2?	Pre-registered gate ≥0.30 GO; published foundation gain ≈ +0.05–0.15 → max landing at ≈ 0.17, below BORDERLINE 0.20. Foundation retry retained as re-entry condition #4.
Q2	Why didn't you do GigaTIME-style multimodal AI?	Different task class (regression vs translation) at different scale (3.2K vs 40M paired cells). Our 8-gene RNA axis = molecular ground-truth target a future thyroid multimodal training would need; complementary, not competing.
Q3	Was hires Visium resolution sufficient?	tissue_hires_scalef ≈ 0.56, 224 px tile ≈ 110 µm — adequate for tissue architecture, insufficient for nuclear morphometry. Resolution explicit re-entry condition #1.
Q4	Slide-level aggregation rescue?	top-25%-mean Spearman = +0.209 at n = 16 — too few for significance; reported as exploratory, not used to flip verdict.
Q5	Other tile sizes tested?	448 px Spearman = 0.017 (worse than 224's 0.022); pre-registered Δr ≥ +0.05 gate not met → 672 cancelled per pre-registration.
Q6	9-dim RGB beats 2,048-dim ResNet50 on raw?	RGB captures stain↔depth color artifact (DM1_raw ~ log_counts ρ = −0.45). Both fall to near-zero on residualized target, validating residualization design.
Q7	Use GigaTIME's released model directly?	Model weights NOT publicly located (GitHub microsoft/ GigaTIME 404; HuggingFace empty as of 2026-05-04). Microsoft typically releases code 6–12 months post-pub.
Q8	ATC AUROC 0.689 contradicts negative?	No. ATC's gross anaplastic morphology ImageNet-reachable; consistent with H&E captures gross histology but not within-tumor molecular gradients (stage ordinal r = −0.05 also fails).
Q9	–	*voice-protected, author keyboard*

13. Anchor files

FINAL DECISION

`CLOSURE_BATTERY_2026_05_04.md` (root, 1-page)

FORENSIC BATTERY

`project/reports/`
`pathology_dm1_closure_battery_2026_0`
(5-failure-mode)

PHASE A BASELINE

`PHASE_A_NOGO_REPORT_2026_05_04.md`

RESULT DATA

`project/results/03_pathology_poc/`
(closure_battery_metrics, negative_controls_raw_
tile_metadata, predictions)

GIGATIME MANUSCRIPT-USE-ONLY

`project/reports/`
`2026_05_04_gigatime_manuscript_use_only.md`
(commit `a1fcac3`)

GIGATIME 6-STRATEGY BUNDLE

commit `df54927` — long-form S1–S6

Cross-paper bridges: Paper 7 (v18 agentic_research framework) Pattern 4 = "Pre-registered failed hypothesis as methods supplement" 의 case study source. Paper 1 + Paper 2B Methods Supplement 에 negative-feasibility 1-pager.

14. 추가 데이터 표 (closure battery raw)

14.1 5-Phase closure battery — 측정값 풀버전

PHASE	TEST	SPEARMAN R	AUROC	RANDOM MAX	BOUND
Phase A	raw DM1 / RAI / TDS @ 224 px ResNet50	0.022 ~ 0.067	0.511 ~ 0.533	0.062	real ≤ random
Phase B	5 residualization (log_counts / log_ngenes / no_resid / rank / epi)	≤ 0.06	–	–	모든 variant 약함
Phase C	Tile size 448 px sweep	0.017	–	–	worse than 224 (672 cancelled)
Phase D	stage ordinal (1 → 4)	-0.05	–	–	random
Phase D	ATC binary	–	0.689	–	BORDERLINE (anaplastic gross morphology)
Phase D	epithelial fraction raw	0.04 ~ 0.08	–	–	weak
Phase E	slide aggregation top25_mean DM1_resid	0.209	–	–	floor 못 넘김
Phase F	simple RGB 9-dim baseline (DM1 raw)	0.224	0.702	–	**stain↔depth artifact** (생물학 신호 X)
Phase G	HF foundation model (UNI / CONCH / Virchow2)	blocked	blocked	–	+0.05–0.15 projected gain (불충분)
neg ctrl	30 random panels (resid)	0.005 ~ 0.031	–	0.031	real ≤ random
neg ctrl	50 random panels (raw)	0.006 ~ 0.062	–	0.062	real ≤ random
neg ctrl	housekeeping_raw (8 genes)	0.013	0.494	–	random level

14.2 Foundation model 후보 + projected gain (시도 못 함)

MODEL	SOURCE	DOMAIN	PRE-TRAIN	PROJECTED DM1_RESID GAIN	이유 못 시도
UNI	HF / Mahmoodlab	Pathology general	100K WSI	~+0.10	HF gated access blocked
CONCH	HF / Mahmoodlab	Vision-language pathology	1.17M image-text	~+0.12	HF gated access blocked
Virchow2	HF / Paige	Pathology general	3.1M WSI	~+0.15	HF gated access blocked
Phikon	HF / Owkin	Pathology histology	40M tile	~+0.05	RunPod stuck pulling image

GO threshold = +0.28 (필요 r ≥ 0.30 또는 AUROC ≥ 0.70). 모든 후보 model projection 미달 → upgrade 무효.

14.3 16-fold LOSO per-slide DM1_resid Spearman + AUROC (Phase A 224 px ResNet50 Ridge)

HELD-OUT SLIDE	STAGE	N_TRAIN	N_TEST	SPEARMAN R	AUROC (DM1_HIGH Q75 GLOBAL)	NOTE
GSM7980860_N-1	PT	3,000	200	+0.005	0.520	baseline; chance
GSM7980861_N-2	PT	3,000	200	−0.054	0.479	worst-2 fold
GSM7980862_N-3	PT	3,000	200	−0.019	0.491	–
GSM7980863_N-4	PT	3,000	200	+0.015	0.545	–
GSM7980864_PTC-1	PTC	3,000	200	**+0.118**	**0.583**	best fold
GSM7980865_PTC-2	PTC	3,000	200	+0.035	0.478	–
GSM7980866_PTC-3	PTC	3,000	200	+0.050	0.542	–
GSM7980867_PTC-4	PTC	3,000	200	**+0.118**	0.520	tied best
GSM7980868_LPTC-1	LPTC	3,000	200	+0.055	0.500	–
GSM7980869_LPTC-2	LPTC	3,000	200	+0.035	0.504	–
GSM7980870_LPTC-3	LPTC	3,000	200	−0.044	0.464	worst AUROC
GSM7980871_LPTC-4	LPTC	3,000	200	−0.024	0.481	–
GSM7980872_ATC-1	ATC	3,000	200	+0.081	NaN	single-class fold
GSM7980873_ATC-2	ATC	3,000	200	+0.002	NaN	single-class fold
GSM7980874_ATC-3	ATC	3,000	200	**−0.074**	NaN	worst Spearman
GSM7980875_ATC-4	ATC	3,000	200	+0.044	0.501	–

Median Spearman = +0.025 · range [−0.074, +0.118] · 16/16 fold ≤ 0.12 (noise level). Median AUROC = 0.501. ATC slides의 NaN AUROC = 모든 ATC tile이 train q75 위 (single-class held-out).

14.4 Negative control panel composition (resid mode, n=30 random + housekeeping)

PANEL TYPE	N PANELS	GENES	DETECTION CRITERION	MEAN SPEARMAN R	MAX SPEARMAN R
Real DM1_like_score_resid	1	TPO, DIO1, TSHR, PAX8, TG, FOXE1, NKX2-1, SLC5A5 (driver-excluded; depth-residualized)	RAI panel	+0.022	+0.022
Housekeeping (resid mode)	1	ACTB, GAPDH, B2M, HPRT1, PPIA, RPL13A, RPLP0, TBP	≥30% spots in all samples	−0.008	−0.008
Random 8-gene (resid mode)	30	random 8-gene panels drawn from genes ≥30% detection rate (excluding RAI_8 + housekeeping)	≥30%	+0.002	+0.031
RAI_8 (raw mode, sanity)	1	same 8 genes, NO residualization	—	+0.067	+0.067
Random 8-gene (raw mode, n=50)	50	random 8-gene from ≥30% detection	—	+0.019	+0.062

Conclusion: real DM1_resid (0.022) ≤ random max (0.031). Real RAI raw (0.067) ≈ random raw max (0.062). Real ≤ noise floor in both modes — closure battery NO-GO 결정적 evidence.

14.5 Phase F simple-RGB feature decomposition (9-dim baseline)

FEATURE	PER-TILE STATISTIC	CHANNEL	RANGE	WHY EFFECTIVE ON RAW TARGET
R_mean	mean	Red	[0, 1]	H&E darkness ↔ sequencing depth proxy
G_mean	mean	Green	[0, 1]	same
B_mean	mean	Blue	[0, 1]	same
R_std	std dev	Red	[0, 0.5]	tissue heterogeneity proxy
G_std	std dev	Green	[0, 0.5]	same
B_std	std dev	Blue	[0, 0.5]	same
R_entropy	Shannon entropy (16 bins)	Red	[0, 4]	texture complexity
G_entropy	Shannon entropy (16 bins)	Green	[0, 4]	same
B_entropy	Shannon entropy (16 bins)	Blue	[0, 4]	same

9-dim total · DM1_raw Spearman r = 0.224 / AUROC 0.702 (>>2,048-dim ResNet50 0.061/0.521). 색상 통계만으로 raw signal capture → stain↔depth artifact 증명. DM1_resid 에선 0.080 (resid 잘 work 증거).

14.6 Closure battery raw data archive

- `project/results/03_pathology_poc/closure_battery_metrics.tsv` — 5-phase metrics
- `project/results/03_pathology_poc/negative_controls_raw_summary.tsv` — 50 random panel raw mode
- `project/results/03_pathology_poc/loso_predictions_resnet50.tsv.gz` — per-spot predictions
- `project/results/03_pathology_poc/tile_metadata{,_resid,_size448,_size672}.tsv.gz` — tile-level metadata

- `project/results/03_pathology_poc/embeddings_resnet50_{224,448}.npz` — embeddings (gitignored, regen via embed_resnet50.py)

15. Future plan (re-entry 5/5 conditions 충족 시)

현재 0/5. 5 condition 모두 충족 시 only.

15.1 Re-entry 핵심

1. **Full-resolution scanner WSI cohort** + paired bulk RNA-seq (40× / 80×) – Visium 아닌 진정한 WSI
2. **External molecular labels** (not ST surrogate) – RNA-seq matched bulk label
3. **Post-marathon explicit decision** (after Paper 1 bioRxiv 6/13)
4. **Foundation model API access** (UNI / CONCH / Virchow2) acquired + **non-thyroid 벤치 검증**
5. **Pre-registered hypothesis** at OSF / AsPredicted

15.2 Methods supplement plan (post-marathon, conditional on re-entry)

1. **Multi-resolution aggregation** – attention-pooling / MIL / multi-scale fusion
2. **Spot-neighbor smoothing** – spatial autoregressive label denoising
3. **Per-slide vs cross-slide normalization variants**
4. **Co-publish with Paper 7** – Pattern 4 case study

15.3 Even if NEVER re-tried — long-term value

- Methods supplement 1-pager → 다음 스튜디오에 actionable upper bound
- 5-failure-mode dissection 자체가 negative-result publication 의 모범
- Paper 7 Pattern 4 의 archetype 으로 영구 보존

16. 한계 · Claim boundary

16.1 이 연구의 한계 (구체적, 솔직하게)

1. **단일 cohort 만 테스트.** GSE250521 (28 slides) + GSE230424 (4 slides) + GSE248205 (8 slides) = 모두 Visium hires (~50 µm spot). Scanner-level WSI (40× / 80×) 는 시도 못함 (TCGA WSI 다운로드 marathon 동안 차단). 따라서 결론은 **"Visium hires 에서"** 의 한정.
2. **Foundation model (UNI / CONCH / Virchow2) 직접 시도 못함.** HF gated access blocked + RunPod stuck pulling image. Projected gain (+0.05 ~ +0.15) 은 publication baseline 에서 추정 — 직접 실측 아님. UNI 가 +0.20 까지 갈 가능성도 배제 못 함 (단, 그래도 GO threshold +0.28 미달).
3. **Residualization variants 5개 만.** log_counts / log_ngenes / no_resid / rank_norm / epi-top50 — 다른 normalization (per-slide standardization / spot-neighbor smoothing 등) 미시도.
4. **Tile size 224 / 448 / 672 만.** 더 큰 tile (1024 / 2048) 또는 multi-scale fusion 미시도. 단 spot 자체가 ~50 µm 라 큰 tile 은 spot 외 영역 포함 = label noise.
5. **Aggregation strategy 3개 만.** mean / top25%mean / highrisk_frac — slide-level top25 0.209 가 가장 나음. attention-pooling / MIL 미시도.
6. **"Pre-registered failed hypothesis" 자체가 unpublished.** 이 NO-GO 가 manuscript supplement 으로만 박힐 예정 — 별도 negative-result paper 로 못 publish (single-cohort, single-task 라 venue 부족).

16.2 Claim boundary table

CLAIM	STATUS
5-failure-mode closure battery executed	YES
Verdict D (full 폐기) at hires Visium	YES
Methods supplement asset (1-pager)	YES
Image-DM1 active pillar	DROPPED from Paper 1/2
Foundation model direct test	BLOCKED (HF gated + RunPod)
TCGA WSI Phase C external validation	BLOCKED (~500 GB)
Re-entry (5/5 conditions)	0/5 — possible after marathon

17. 데이터 / 다운로드

📥 **CLOSURE_BATTERY_METRICS.TSV**
5-phase × 8-test full metrics

📥 **NEGATIVE_CONTROLS_SUMMARY.TSV**
30 random panels (resid mode) + housekeeping

📥 **NEGATIVE_CONTROLS_RAW**
50 random panels (raw mo

📥 **LOSO_METRICS_RESNET50.TSV**
LOSO ResNet50 224 px metrics per slide

→ **PAPER 2B (HT-OVERLAP)**
Pivot destination — molecular-only mechanism

→ **PAPER 7 FRAMEWORK**
Pattern 4 = pre-registered f
hypothesis

← **PAPER 1 (DM1 MOLECULAR)**
Methods supplement parent

← **HUB INDEX**
8-papers portfolio

Status. Paper 2A = NO-GO 확정 2026-05-04 · Methods supplement asset · "Failure 가 아니라 finding" — actionable upper bound 를 다음 스튜디오에 제공.

Live URL. `http://40.82.129.113:8012/papers_hub_2026_05_04/paper2a.html`